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# QINGYI LI

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## RESEARCH INTERESTS

Individualized brain topography · Precision functional mapping · Gene–environment interplay · Brain dynamics
Computational modeling · Internalizing psychopathology · Risk and resilience · Adolescent development

## EDUCATION

**Virginia Tech,** Blacksburg, USA — Aug 2025 – Present
*Ph.D. Student, Developmental Science, Department of Psychology*
*Advisor: Prof. Tae-Ho Lee*

**Hangzhou Normal University,** Hangzhou, China — Sep 2021 – Jul 2024
*M.A. in Applied Psychology*
*Advisor: Prof. Bao-ming Li*

**The University of British Columbia,** Vancouver, Canada — Jun 2023 – Aug 2023
*Visiting Master's Student, Overseas Study Program, Faculty of Education*

**Henan University,** Kaifeng, China — Sep 2017 – Jul 2021
*B.S. in Applied Psychology*

## RESEARCH GRANTS

**Seed Grant Program for Scientific Research, Hangzhou Normal University** — 2022–2023
Role: Principal Investigator
Title: A Questionnaire and Brain Imaging Study on the Relationship Between Stressful Life Events and Sleep in University Students

**Pilot Study, STI2030-Major Projects (China Brain Project)** — 2022–2023
Role: Pilot Study Lead
Title: Cognitive, Emotional, and Intellectual Profiles of Children with Long-term Weiqi (Go) Training

**General Research Project, Zhejiang Provincial Department of Education** — 2022–2024
Role: Co-Investigator
Title: Effects of Computerized Working Memory Training on Individual Proactive and Reactive Control

## PUBLICATIONS

*Peer-Reviewed Journal Articles*

[1] **Li, Q.**, Ye, X., Li, Z., Yang, S., Yin, L., Li, B.-M., & Wang, C. (2024). Association between stressful life events and sleep quality in Chinese university students: Mediating and moderating roles of emotion regulation. *Journal of Health Psychology*, *30*(5), 909–920. [DOI]

[2] Ren, T., Ye, X., Li, Z., **Li, Q.**, Zhang, X., Dou, W., Jia, X., Li, B.-M., & Wang, C. (2023). Associations between physical activity and proactive control and the modulating role of working memory. *Psychology of Sport and Exercise*, *66*, 102374. [DOI]

*Manuscripts Under Review*

[3] **Li, Q.*,** Ye, Y.*, Ren, T., Xue, M., Lee, T.-H., & Li, B.-M. (2026). Cumulative stress contextualizes associations between vmPFC–limbic connectivity and sleep quality across psychological resilience profiles [Under review at *Biological Psychology*, special issue "Resilience in the Biological–Psychological–Social Cascade"]. *Equal contribution.

[4] **Li, Q.**, Chen, Y.-Y., Kim-Spoon, J., Casas, B., Qu, Y., & Lee, T.-H. (2026). Parent–child cortical similarity indexes multiple dimensions of sensitivity in the association between family environment and adolescent internalizing symptoms [Under review]. *bioRxiv*. [DOI]

[5] **Li, Q.**, Chen, Y.-Y., Zhou, Z., Qu, Y., Kim-Spoon, J., & Lee, T.-H. (2026). A convergent behavioral–neural profile of parental dysregulation and the link between parent–child brain-state similarity and child negative affect [Under review]. *bioRxiv*. [DOI]

[6] Lee, T.-H., Chen, Y.-Y., **Li, Q.**, Yang, B., Zhou, Z., & Qu, Y. (2026). Family conflict and parent–child similarity in affective valuation [Under review]. *bioRxiv*. [DOI]

[7] Lv, X., Wu, Z., Zhang, Z., Xue, M., Guo, J., Liu, S., **Li, Q.**, Liu, B., He, X., Xi, Y., Wang, K., Chen, Z., & Chen, R. (2026). A multimodal neuroimaging large language model for transdiagnostic suicidal risk assessment across developmental stages [Under review at *Nature Machine Intelligence*].

*Manuscripts in Preparation*

[8] **Li, Q.**, et al. (in preparation). Baseline levels of daily affect, not affective dynamics, predict suicidal ideation in college students two months later: Evidence from hidden Markov models and external validation

[9] **Li, Q.**, et al. (in preparation). Partitioning a child's individualized network topography into species-shared, parent-shared, and child-specific parts: Evidence from precision functional mapping of parent–child dyads.

[10] Chen, Y.-Y., Kim, H., **Li, Q.**, & Lee, T.-H. (in preparation). Shared and distinct neural responses during aggressive and prosocial behaviors: A systematic review of fMRI studies across adolescence and young adulthood.

## CONFERENCE PRESENTATIONS

[11] **Li, Q.**, Chen, Y.-Y., Zhou, Z., Qu, Y., Kim-Spoon, J., & Lee, T.-H. (2027, April). *Parent–child neural similarity in the link between family environment and adolescent adjustment* [Symposium paper proposal submitted]. Society for Research in Child Development Biennial Meeting, Atlanta, GA, USA.

[12] **Li, Q.**, Chen, Y.-Y., & Lee, T.-H. (2026, November). *Parent internalizing shapes the link between parent–child latent brain-state similarity and child recent negative affect* [Nanosymposium talk]. Society for Neuroscience Annual Meeting, Washington, DC, USA. (Trainee Professional Development Award)

[13] **Li, Q.**, Chen, Y.-Y., & Lee, T.-H. (2026, March). *A topological principle of parent–child brain morphological similarity* [Poster and Data Blitz presentation]. Cognitive Neuroscience Society Annual Meeting, Vancouver, BC, Canada. (Graduate Student Award)

[14] **Li, Q.**, Zhang, Y., Li, B.-M., & Li, Z. (2023, October). *The effect of expectation on the regulatory mechanisms of cognitive control* [Oral presentation]. 25th National Academic Congress of Psychology, Chinese Psychological Society, Chengdu, China.

[15] Yang, J., Zhang, Y., **Li, Q.**, Li, B.-M., & Li, Z. (2023, October). *The influence of information fuzziness and group pressure on conformity behavior and its neural mechanism* [Oral presentation]. 25th National Academic Congress of Psychology, Chinese Psychological Society, Chengdu, China.

[16] **Li, Q.**, Ye, X., Li, B.-M., & Wang, C. (2023, October). *The mediating role of emotional regulation in the associations of stressful life events with sleep quality and depressive experience in Chinese university students* [Poster presentation]. 25th National Academic Congress of Psychology, Chinese Psychological Society, Chengdu, China.

## RESEARCH EXPERIENCE

*Note. Numbers in brackets refer to entries in Publications and Conference Presentations.*

**Graduate Researcher, Affective Neuroscience and Development Lab, Virginia Tech** **Aug 2025 – Present**

*Advisor: Prof. Tae-Ho Lee*

- **Lead project · Cortical morphometric similarity**: Asked whether sensitivity to family environment resides in the parent–child relationship rather than in the child alone; separated regional from organizational cortical similarity (400-parcel morphometry, morphometric similarity networks; 115 dyads, two cohorts); conceived the question and led the analysis and writing [4, 11, 13].

- **Lead project · Brain-state dynamics (HMM)**: Tested whether parent–child brain-state similarity is protective or risky depending on the parent; modeled time-resolved brain states during an emotional film (HMM-MAR, neural index of parental state dynamics; 32 dyads); led the analysis and writing [5, 12].
- **Lead project · Precision functional mapping**: Set out to resolve where, how much, and in what unit genes and environment are jointly passed from parent to child in the brain; partitioned each child's network topography into species-shared, parent-shared, and child-specific parts (MS-HBM, variance partitioning, dyadic identifiability; 28 dyads); conceived the question and led the design, analysis, and writing [9].
- **Pipeline lead · 8-year longitudinal adolescent morphometry**: Asked how cortical structure changes across adolescence; built a FreeSurfer longitudinal pipeline extracting thickness, surface area, curvature, and gyrification across waves (with Prof. Jungmeen Kim-Spoon); led pipeline development [4].
- **Side projects · Systematic review and manuscript editing**: Co-authored a systematic review on aggressive and prosocial behavior; edited a manuscript on family conflict and parent–child affective valuation [6, 10].

**Collaborative Project (Remote), University of Notre Dame** **Jun 2026 – Present**

*Precision Brain Mapping Lab; Supervised by Dr. Jingnan Du*

- **Pipeline lead · Precision functional mapping at 7T**: Deployed the iProc pipeline (Buckner Lab, Harvard) on Notre Dame's computing cluster and applied it to densely sampled 7T data (Natural Scenes Dataset); led the deployment.
- **Lead analyst · Individualized language networks**: Asked whether two nested language networks replicate in an independent 7T dataset; estimated individual networks with MS-HBM and verified boundaries with seed-based connectivity; led the analysis.
- **Lead analyst · Cortico-subcortical mapping**: Asked whether the cortical dissociation holds in cerebellum, thalamus, and striatum; mapped subcortical voxels to individualized cortical networks; led the analysis.
- **Method transfer · Parent–child dyadic precision functional mapping**: Adapted the individual-level mapping workflow for parent–child dyads [9].

**Collaborative Project (Remote): E-MIND, Multi-institutional** **Sep 2026 – Present**

*Ecological Mechanisms in Young Adults' Mental Health*
*Supervised by Prof. Shufang Sun (Brown University) and Prof. Mark Shuquan Chen (Harvard University); Collaborator: Averi Junkang Lin (Purdue University) and Kaiwen Bi (University of Hong Kong)*

- **Lead analyst · Affect-dynamics modeling (EMA, HMM)**: Examined whether how affect moves adds information beyond where it sits; modeled 34 affect-dynamics indices, including hierarchical HMMs, in a two-university EMA study ($N$ = 2,549); led all analyses.
- **Lead analyst · Prediction and validation (ML)**: Asked whether apparent dynamics reflect genuine signal or artifacts of mean levels; built cross-validated, externally validated prediction models and null simulations; led all analyses.
- **Lead project · Daily affect and suicidal ideation**: Leading a first-author manuscript on which features of daily affect predict later suicidal ideation [8].

**Research Assistant, Youth Mental Health Lab, Tsinghua University** **May 2024 – May 2025**

*Advisors: Prof. Runsen Chen, Dr. Zhijun Wu, & Dr. Diyang Qu*

- **Independent analysis · Multimodal MRI in adolescent self-injury**: Asked how brain structure and connectivity differ in adolescents with NSSI and emotion-regulation difficulties; analyzed gray matter and static and dynamic connectivity (Peking University Sixth Hospital cohort); led the analysis independently.
- **Analyst · Preliminary analyses in large cohorts (A-HELP, ABCD)**: Ran preliminary linear mixed models in the ABCD Study ($N \approx$ 11,800) linking pre-computed polygenic risk scores and subregional gray matter volume to NSSI, and latent profile and network analyses in the A-HELP cohort ($N$ = 94,412; 393 schools) of family environment, emotion regulation, and personality; findings reported to the PI.

- **Co-designer · Neuromodulation (taVNS) study**: Co-designed a cross-site study of transcutaneous auricular vagus nerve stimulation (taVNS) mechanisms in adolescents; drafted the IRB protocol.
- **Pipeline builder · MRI preprocessing**: Built the lab's BIDS-based preprocessing pipeline and ran resting-state connectivity and VBM analyses for 2 MRI datasets.
- **Data collection · Multimodal clinical assessment**: Collected interview, EMA, EEG, and fMRI data and trained six interns [7].

**Master's Student Researcher, Hangzhou Normal University** **Sep 2021 – May 2024**

*Institute of Brain Science; Advisors: Prof. Bao-ming Li, Prof. Chunjie Wang, Dr. Zhenghan Li, & Dr. Zhong Yang*

- **Grant PI · Stress, sleep, and resting-state fMRI**: Led a questionnaire and MRI study of stressful life events and sleep as PI of a university grant: collected longitudinal survey data from 1,169 university students and multimodal MRI (T1, resting-state, DTI) from 69 participants; tested mediating and moderating roles of emotion regulation and ROI-based amygdala and medial prefrontal connectivity [1, 3, 16].
- **Pilot lead · Long-term Weiqi training in children**: Led a pilot study on cognitive, emotional, and intellectual profiles of children with long-term Weiqi (Go) training; programmed E-Prime tasks (Raven's matrices, Go/No-Go, cued task switching) and assessed 159 children and adolescents of Raven's Progressive Matrices, cognitive tasks and socioemotional questionnaires.
- **Study lead · EEG and eye-tracking**: Designed and ran an EEG and eye-tracking study (Neuroscan, Tobii, E-Prime; $n = 64$) on how expectation and statistical learning regulate attention to targets and distractors [14, 15].
- **Co-Investigator · Working memory training**: Co-investigator on a provincial grant studying proactive and reactive control: reviewed the literature for the grant application, adapted AX-CPT, N-back, and backward digit span tasks, and preprocessed structural and resting-state MRI (SPM, DPABI, CONN) [2].

## TEACHING EXPERIENCE

**Virginia Tech, Department of Psychology** Blacksburg, VA
*Graduate Teaching Assistant, Introduction to Psychology* Spring 2026 – Present
*Graduate Teaching Assistant, Developmental Psychology* Fall 2025

**Tianyuan College** Hangzhou, China
*Intern Mental Health Teacher (Grades 4–6)* Sep 2023 – Dec 2023

**167 PSYCLASS** Hangzhou, China
*Founder and Tutor, Master's Entrance Exam Preparation in Psychology* Sep 2021 – Feb 2022

**Jiying Primary School** Kaifeng, China
*Trainee Mental Health Teacher (Grade 1)* Sep 2020 – Dec 2020

## MENTORING AND SERVICE

**Graduate Application Mentor (Pro Bono)**
*PsyCN2none (Team Lead)* 2025–Present
*1HourProject* Fall 2025
*NPSP* Fall 2025

- Provide one-on-one mentoring to students applying to graduate programs in psychology, including program selection and review of statements of purpose, personal statements, and CVs.
- Coordinate a volunteer team of 17 mentors supporting 10 mentees: organize application workshops, set the timeline, and track progress and feedback across the application cycle.

**"Right to Play" (NGO), Kaifeng Children's Hospital** Kaifeng, China
*Part-time Play Coach* Oct 2018 – Dec 2019

- Used therapeutic play and picture-book preparation to reduce distress in chronically ill children and children awaiting surgery.

## SKILLS

**Statistical Methods:** Multiple regression, mediation and moderation analysis, structural equation modeling, cross-lagged panel models, growth modeling, latent class/profile analysis, network analysis, linear mixed and multilevel models, hidden Markov models (including hierarchical HMM), intensive longitudinal (EMA) modeling of affect dynamics, matched case–control designs

**Machine Learning:** Penalized regression, gradient boosting, cross-validation, external validation, calibration, decision-curve analysis, connectome-based predictive modeling

**Neuroimaging Methods:** Task activation; static and dynamic functional connectivity; seed-based connectivity; graph-theoretic network analysis; effective connectivity; HMM-MAR brain-state analysis; voxel- and surface-based morphometry; morphometric similarity; precision functional mapping (individualized parcellation, network variants, variance partitioning); cortico-subcortical mapping

**Neuroimaging Software:** FreeSurfer, fMRIPrep, Connectome Workbench, SPM, CAT12, DPABI, CONN

**Programming and Statistics:** R, MATLAB, Mplus, SPSS; Linux high-performance computing clusters

**Data Collection:** E-Prime; eye tracking (Tobii); EEG (Neuroscan, Brain Products, Curry 8); MRI (GE Discovery MR750 3T, Siemens Prisma 3T); EMA; semi-structured clinical interviews

**Languages:** Chinese (native), English (proficient)

**Certification:** Teacher Qualification Certificate, Senior High School Psychology (China)

## SCHOLARSHIPS AND AWARDS

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| Trainee Professional Development Award, Society for Neuroscience (USD 1,000) | 2026 |
| Graduate Student Award, Cognitive Neuroscience Society (USD 100) | 2026 |
| Academic Scholarship, Hangzhou Normal University (CNY 9,000) | 2021–2023 |
| School Outstanding Student Cadre, Hangzhou Normal University (CNY 1,000) | 2023 |
| Prize for Community Work, Hangzhou Normal University (CNY 1,000) | 2022–2023 |
| Excellent Graduate Student, Hangzhou Normal University (CNY 1,000) | 2022–2023 |
| Practical Service Award, Hangzhou Normal University (CNY 1,000) | 2022–2023 |
| Exchange Student Fellowship, Hangzhou Normal University (CNY 120,000) | 2023 |
| Second Prize, National Applied Psychological Practice Skills Competition, Hangzhou Normal University | 2022 |
| Third Prize, "Challenge Cup" National College Student Curricular Academic Science and Technology Works Competition, Henan University | 2020 |

## MEMBERSHIPS

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| Society for Neuroscience | 2026–Present |
| Society for Research in Child Development | 2026–Present |
| Cognitive Neuroscience Society | 2025–Present |
| Association of Chinese Helping Professionals and Psychologists–International | 2024–Present |
| Chinese Psychological Society | 2022–Present |
| Organization for Human Brain Mapping | 2021–2022 |